



FRESNO STATE · CSCI

USHER × FINISH IN FIVE

Four problems that connect my research

From mathematical structure to collective intelligence

THEORY

1

SHAPE

Geometry &
topology

BRIDGE

2

SCIENCE

Theory meets
application

APPLICATION

3

WORLDS

3D & video
generation

PRIORITY

4

TEAMS

Agentic AI &
networks

RECENT: QUANTUM ALGORITHMS

Cheng Xin · Assistant Professor of Computer Science



How can AI understand the shape of data?

Geometric & Topological Machine Learning

Complex data may live on graphs, curved spaces, or evolving networks. The goal is to preserve the structure that carries meaning.

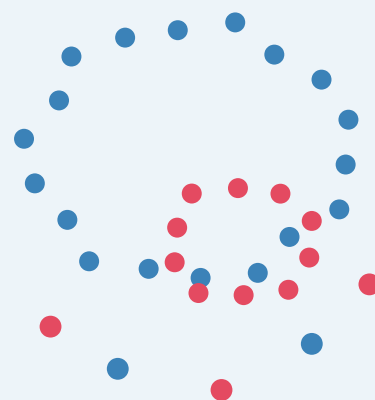
STABLE

INTERPRETABLE

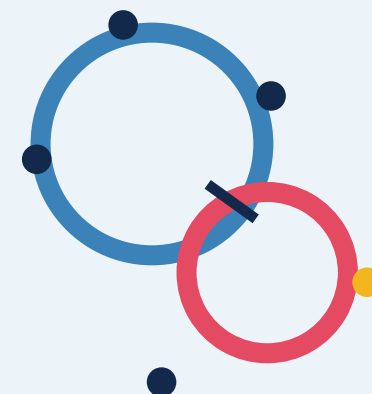
EFFICIENT

Preserve what matters—even when the data are noisy.

FROM POINTS



TO SHAPE



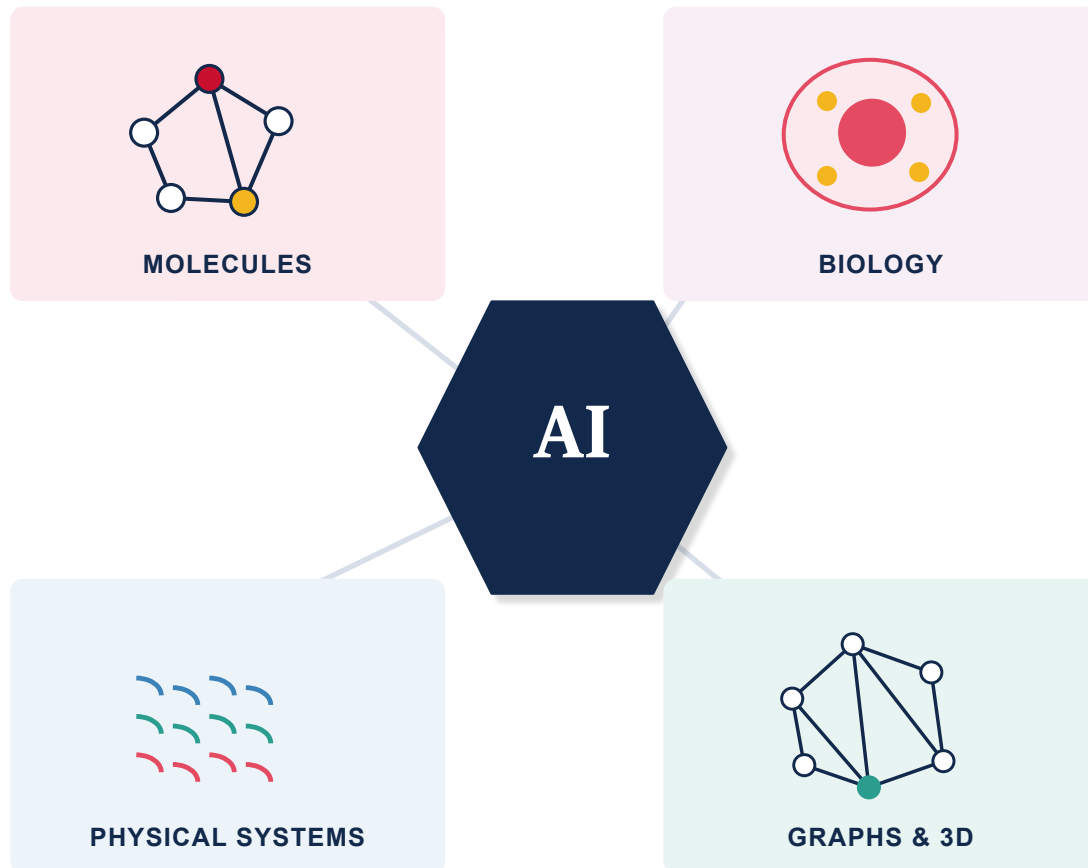
connections · loops · global organization

RECENT QUANTUM INTEREST

Can quantum computing find structure faster?



SCIENCE HAS STRUCTURE



How can AI learn from science—and respect its rules?

AI for Science

Use geometry, topology, and domain knowledge as guides—not afterthoughts.

PATTERNS

+

PRINCIPLES

Accurate is good. Scientifically meaningful is better.

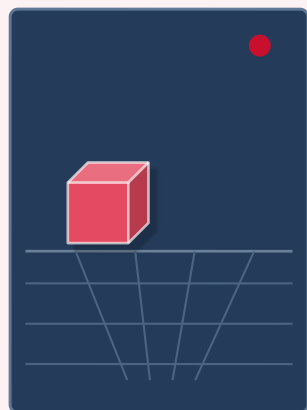
Can generated worlds stay consistent?

APPLICATION · 03

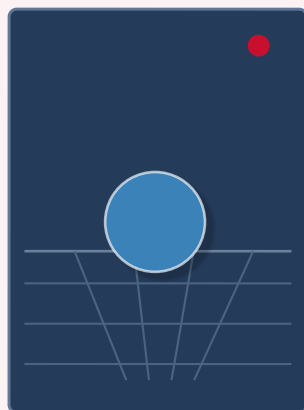
3D & Video Generative Models



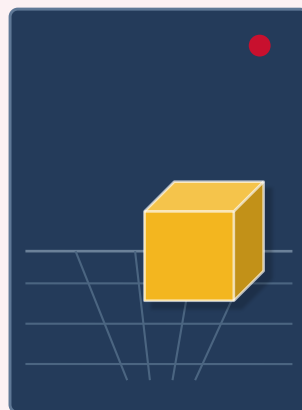
FRAME QUALITY ONLY



t1



t2

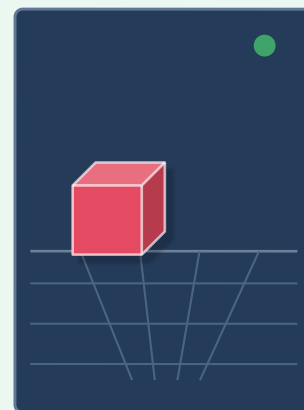


t3

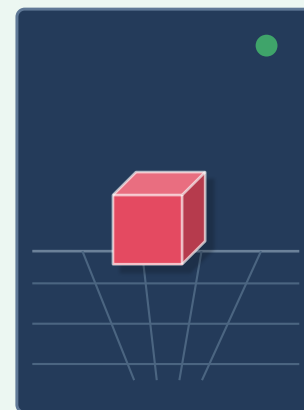
A beautiful frame can still drift, morph, or break.



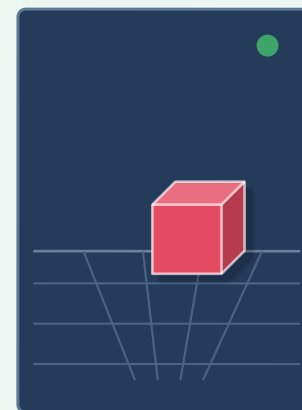
A CONSISTENT WORLD



t1



t2



t3

Geometry, viewpoint, and motion agree across time.

DATASETS

BENCHMARKS

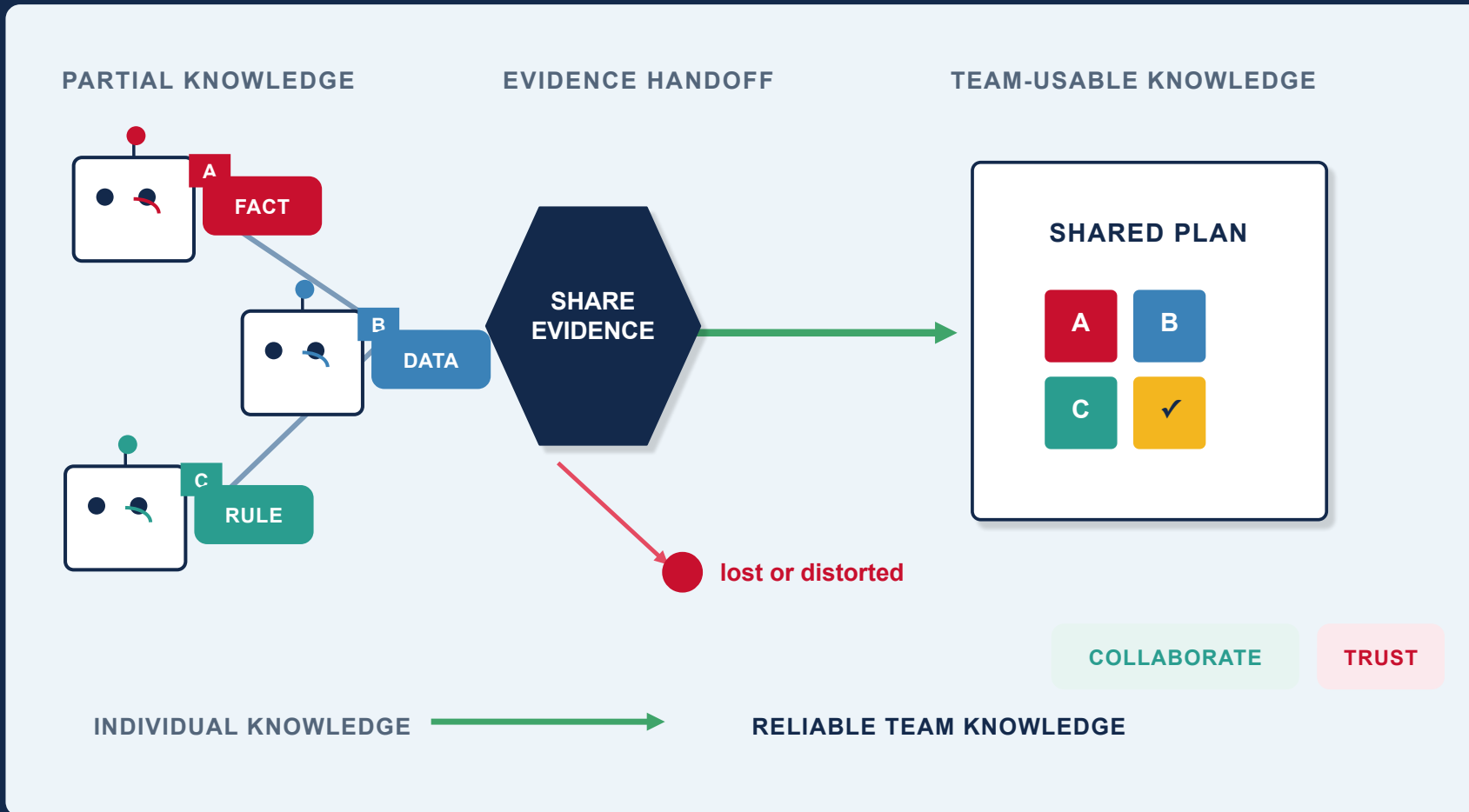
EVALUATION

Quality is not enough: the content should remain the same world.

Can many AI agents become one reliable team?

CURRENT PRIORITY · 04

Agentic AI · Multi-Agent Collaboration & Networks



QUESTIONS?



Explore projects &
papers

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Thank you